Tmaxius

Equipment Engineer → Automation Specialist → Software Developer

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PROFESSIONAL SUMMARY

A versatile engineer-turned-developer with 10 years of hands-on experience at AMD's manufacturing facility, progressing from Electronic Technician to Equipment Engineer. Combines deep hardware expertise with self-taught software development skills to build practical automation and consumer applications. Proven ability to identify operational inefficiencies and deliver custom solutions — from equipment automation to POS systems, photo kiosks, and metaphysical / astrological apps.

CORE COMPETENCIES

- Equipment Engineering & Automation
- Electronics & Mechanical-Electronic Integration
- Yield Improvement & Process Optimization
- POS System Development
- Kiosk Application Development
- Mobile / Web App Development
- Numerology & BaZi (Chinese Astrology) Systems
- Hardware–Software Integration

PROFESSIONAL EXPERIENCE

AMD — Manufacturing Facility

10 Years

Electronic Technician → Equipment Engineer

- Started as Electronic Technician, gaining foundational expertise in semiconductor manufacturing equipment.
- Promoted to Equipment Engineer based on technical proficiency and problem-solving capabilities.
- Maintained, troubleshot, and optimized complex electronic and mechanical systems.
- Collaborated with cross-functional teams to ensure equipment reliability and production continuity.

KEY ACHIEVEMENTS

- Enhanced mechanical-electronic operations to improve equipment performance.
- Implemented process improvements that increased product yield.
- Developed deep understanding of equipment lifecycle and maintenance optimization.

Equipment Automation

Post-AMD

Automation Specialist

- Involved in equipment automation initiatives, bridging hardware and software systems.
- Applied engineering background to design and implement automated solutions.
- Focused on reducing manual intervention and improving operational efficiency.

SOFTWARE DEVELOPMENT PROJECTS

POS System — Cash Register Retrofit

SELF-TAUGHT DEVELOPER

Modified and retrofitted cash register hardware; self-learned software development to build a custom Point-of-Sale system. Delivered a functional, tailored POS solution from hardware modification through software implementation.

PhotoKiosk — Photo Shop Application

DEVELOPER

Developed a photo kiosk application for photo shops. Enabled customers to easily select, edit and print photos; created an intuitive user interface that streamlines the selection and printing workflow.

Numerology Philosophy App

DEVELOPER

Studied numerology philosophy independently and built a mobile application where users input their date of birth to generate a personalised character analysis, with V-chart visualisation and charted results.

BaZi (Chinese Astrology) App

DEVELOPER

Studied BaZi (Four Pillars of Destiny) under a teacher using classical BaZi texts, then applied those principles to build a BaZi analysis application — translating traditional astrological calculations into a functional digital tool.

R&D concepts (not yet shipped): AI facial-diagnosis app · mechanical mobility leg for amputees and the elderly — see projects.html.

TECHNICAL SKILLS

HARDWARE	Electronic systems · mechanical-electronic integration · equipment maintenance
AUTOMATION	Equipment automation · process optimization · yield improvement
SOFTWARE	POS development · kiosk applications · mobile / web app development (JavaScript, React, Node.js, PHP, MySQL, Java, Flutter)
DOMAIN KNOWLEDGE	Numerology · BaZi astrology · semiconductor equipment engineering

EDUCATION & SELF-DEVELOPMENT

- **Electronic Technician Training** — foundational electronics
- **On-the-job Engineering Training** — AMD equipment engineering
- **Self-Taught Software Development** — POS, kiosk and app development
- **Numerology Philosophy** — independent study
- **BaZi (Four Pillars of Destiny)** — studied under a teacher with classical texts

CAREER HIGHLIGHTS

- ✓ 10 years of progressive experience at a leading semiconductor company.
- ✓ Successfully transitioned from hardware engineering to software development.
- ✓ Built 4+ functional applications across diverse domains (POS, photo kiosk, numerology, BaZi).
- ✓ Self-directed learner who converts domain knowledge into working software products.